Daniel Scalena

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Education

Double Doctorate (Ph.D) in Computer Science – NLP *Nov 2023 – exp. Q4 2026*
University of Milano - Bicocca | University of Groningen (CLCG)

Research: I study how language models form, represent, and act on information, using interpretability to make their reasoning and behaviour safer and more reliable.

Mediterranean Machine Learning (M2L) Summer School *Sep 2024*
Milano, Italy

1-week program led by Google DeepMind scientists; presented my paper on multi-property LLM steering.

M.Sc in Computer Science, ML track *Oct 2021 – Oct 2023*
University of Milano - Bicocca

- Final Grade: 110/110 with honors, (GPA: 29.7/30)
- Thesis: On the explainability of Large Language Models detoxification.

B.Sc in Computer Science *Oct 2018 – Jul 2021*
University of Milano - Bicocca

- Final Grade: 110/110 with honors, (GPA: 28.3/30)
- Thesis: Hate speech detection on social networks using state-of-the-art NLP techniques.

Experience

Cohere *June 2026 – Oct 2026*
Intern – Member of Technical Staff

- Led rubric-reward research to shape coding-agent behaviour via RL; first-author paper.
- Built agents-as-a-judge, rubric generation, and rubric-based training/eval from scratch on internal infra.

University of Groningen, Computational Linguistic Group *Apr 2023 – Jul 2023*
Research Intern

- Aligned LMs to counter hate speech narratives using RLAIF, reaching up to 50% in safety improvements;
- Analyzed and interpreted model behavior shifts after applying post-training algorithms.

C.I.N.I. & University of Milano - Bicocca *Jun 2022 – Apr 2023*
Assistant Researcher

- Devised Open Relation Extraction for Italian legal texts, improving previous accuracy by 30%;
- Project funded by the *Italian Ministry of Justice* and *Inter-university Consortium for Computer Science.*

TESTUDO s.r.l. *Jun 2022 – Aug 2022*
ML Engineer (contractor)

- Designed and trained deep learning models to automate the company's timestamp recognition system.

Main publications

Beyond the Commitment Boundary: Probing Epiphenomenal Chain-of-Thought in Large Reasoning Models
Daniel Scalena*, Sara Candussio*, Luca Bortolussi, Elisabetta Fersini, Malvina Nissim, Gabriele Sarti
Preprint – under review ICLR 2027

Demystifying Entropy-based Selection for Chain-of-Thought Compression in Large Reasoning Models
Sara Candussio*, **Daniel Scalena***, Luca Bortolussi, Elisabetta Fersini, Malvina Nissim, Gabriele Sarti
Proceedings at the 9th BlackBoxNLP Workshop, (at EMNLP 2026)

What Makes a Good Rubric Reward? Reinforcement Learning for Coding Agents Beyond Correctness
Daniel Scalena, Leon Engländer, Sarra Habchi, Patrick Lewis, Ahmet Üstün, Sophia Althammer
Preprint

EAGer: Entropy-Aware GEneRation for Adaptive Inference-Time Scaling
Daniel Scalena, Leonidas Zotos, Elisabetta Fersini, Malvina Nissim, Ahmet Üstün
Proceedings of the 43rd International Conference on Machine Learning, (ICML 2026)

Steering Large Language Models for Machine Translation Personalization
Daniel Scalena*, Gabriele Sarti*, Arianna Bisazza, Elisabetta Fersini, Malvina Nissim
Proceedings of the 19th Conference of the European Chapter of the Association for CL, (EACL 2026)

Multi-property steering of large language models with dynamic activation composition
Daniel Scalena, Gabriele Sarti, Malvina Nissim
Proceedings at the 7th BlackBoxNLP Workshop, (at EMNLP 2024)

A gentle push funziona benissimo: making instructed models in Italian via contrastive activation steering
Daniel Scalena, Gabriele Sarti, Malvina Nissim, Elisabetta Fersini
Proceedings at the 10th Italian Conference on Computational Linguistics, (CLIC-it 2024)

Let the Models Respond: Interpreting Language Model Detoxification Through the Lens of Prompt Dependence
Daniel Scalena, Gabriele Sarti, Malvina Nissim, Elisabetta Fersini
Extended abstract at the 6th BlackboxNLP Workshop, (at EMNLP 2023)

Full list of publications

Scholarships and Awards

ICML Top Reviewer (Gold Reviewer) – free conference registration	*2026*
Google and Apple – BlackBoxNLP (EMNLP 2023) travel award	*2023*
Full-Fee M.Sc. Scholarship due to merit and financial scores	*2021 – 2023*
Full-Fee B.Sc. Scholarship due to merit and financial scores	*2018 – 2021*

Projects

Reward LM *DanielSc4/RewardLM*

- Python toolkit library for Fine-Tuning and Reinforcement Learning from AI Feedback of Generative LMs;
- Developed tools to interpret and compare toxicity differences between original and fine-tuned models.

Activities

Co-organiser	Challenge the Abilities of LAnguage Models in ITAlian (CALAMITA) *2024*
Teaching	NLP course labs, MSc Data Science @ UniMiB (*2024 – 2026*)
Reviewer	NeurIPS *2026*, ICLR *2027*, ICML *2026*, EMNLP *2025 – 2026*, BlackBox-NLP *2025 – 2026*, ACL *2025*, LREC-Coling *2024*, NAACL *2024**, Information Processing and Management *2024 – 2025*, CliC-IT *2023 – 2026*, ICML Mechanistic Interpretability Workshop *2024** (*secondary reviewer)
Volunteer	CliC-IT *2021*, EMNLP *2023*
Misc.	European Researchers Night GroNLP collaborator *2023*

Languages

Italian	Native Language
English	C1 – B-between Foreign Languages
French	A2